

Our needs have evolved beyond the basic necessities of food, shelter and clothing. We have developed a number of devices to make our lives simpler and better. We want easy control of devices around us, and do not want to wait in queues to pay toll taxes or utility bills. We want to monitor our health status, track our assets and vehicle movements from a single device.

As technology grows, IoT solutions will change the way we live, work and do business. The IIoT is a combination of hardware, like sensors, connectors and gateways, and application software in a single system. IIoT devices can communicate with each other and also with an IIoT hub, which could be a mobile app or a server. A hub collects data from devices and analyses it to provide meaningful output to users.

Benefits of IIoT solutions for businesses

The tiny sensors can be attached to everything from LED bulbs to self-driving cars. These record and send data to the cloud. This allows collecting specific feedback on how products or equipment are used, their lifespan and future requirement of users. It also helps companies make smarter products, enable smarter business operations and make smarter decisions.

The driving philosophy behind the Industrial IIoT (IIoT) is that, smart machines are good at accurately and consistently capturing and communicating data. This data enables companies detect reasons for inefficiencies and problems sooner, saving both time and money.

Enhanced flexibility, higher speed and increased efficiency in the production process are the objectives of adapting IIoT solutions for businesses. Development and adoption of the IIoT are being driven by multiple factors, including easily available low-cost sensors, increase in bandwidth and processing power, wide usage of smartphones, availability of Big Data analysis tools and scalability of Internet Protocol version 6 (IPv6).

Enterprises are adopting IIoT solutions for their benefits that include reduction in cost, optimisation in operations and improvement in efficiency.

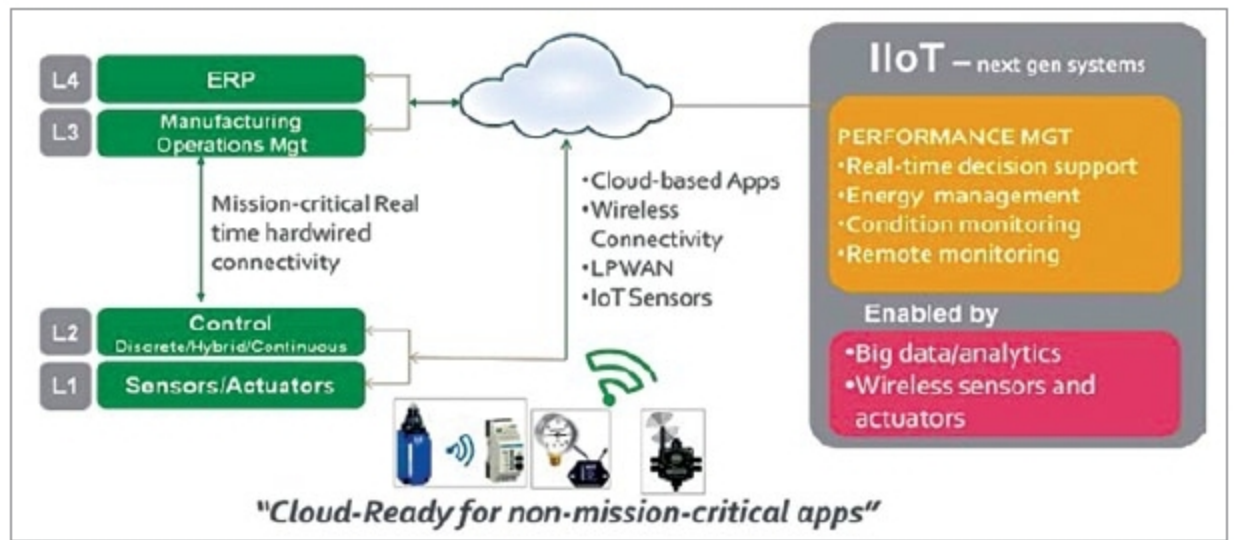


Fig. 1: The next-generation IIoT (Credit: www.mhi.org)

They are now starting to focus on improving services and customer experiences with IIoT-connected solutions.

Impact of the IIoT on the manufacturing industry

In the IIoT, multiple devices are connected by communications software to collect, monitor, exchange, analyse and instantly act on information, to intelligently change their behaviour or their environment, all without human intervention. Connected things in the IIoT provide tremendous opportunities to increase performance and lower cost by leveraging sensor data, machine-to-machine (M2M) communication and automation technologies.

The IIoT has significant potential to improve automation, industrial control, quality control, supply chain traceability, overall efficiency and asset performance. Manufacturing, workflows, supply chains and business processes are becoming more intelligent, flexible and dynamic. These are being improved through self-optimisation and autonomous decision-making by using Big Data analytics and deep learning by predicting failures or misalignments on production lines.

Businesses are increasingly relying on digital twins integrated with advanced analytics to improve operations. Digital twins enable business leaders to predict plant performance, ageing patterns, anomalies and future behaviour. These carry deep domain knowledge of specific industrial assets from which the twins can display models.

Digital twins acquire a massive amount of design, manufacturing, inspection, repair, online sensor and

operational data feeds through continuous monitoring. With various computational physics-based models and advanced analytics, these determine asset health, performance and lifetime.

Digital twins can also run on cloud-based industrial platforms, which are designed to take and analyse machine sensor data. This improves product design and manufacturing with connected real-world sensor data.

Eric Farrugia, vice president - strategic marketing, industrial controls and drives, Schneider Electric, says "The IIoT is blurring the boundaries between physical and virtual objects. It provides more flexible models of accessing process and machine data. Generating and collecting data alone will not enable manufacturers to realise the full benefit of the IIoT. There are other components to consider, such as an open and collaborative environment to allow partners to develop a common platform, smart connected devices to generate information, edge gateway to aggregate data, display information and connect to the cloud for advanced analytics, and applications to further analyse data to help reach business goals."

Developments in wireless technologies empowering the IIoT businesses

Wireless technologies are driving Industry 4.0, taking productivity and efficiency to new levels. As a result, wireless connectivity is becoming even more business- and mission-critical, leading to very strict requirements with respect to reliability, low latency and security. Next-generation wireless technologies such as 5G for ultra-reliable, low-latency communications